Day 13 — Genome Annotation

BIOL 4810

2026-03-08

Day 13 — Genome Annotation + Species Identification

Goal: Use **Prokka** (Galaxy) to annotate a bacterial genome assembly, interpret the output files, extract the **16S rRNA** for a quick BLAST-based ID, and then use **TYGS** + **iTOL** to place your genome on a phylogenomic tree.
Submission: Fill in the answers + upload required screenshots/files below → click **Prepare submission preview** → click **Download as PDF (Print)**.

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0) Files you need today

You should already have (from Day 12): - SPAdes assembly contigs file (FASTA)

0A. Assembly file name (as it appears in Galaxy History):

27: SPAdes on dataset 15 and 16 and collection 12: Contigs

Local folder (for Git later)

Create a folder inside your BIOL 4810 course folder, for example: - `module2_wgs/day13_prokka_tygs/`

0B. Local folder path/description:

Bioinformatics/biol4810/module02/Annotation/

0C) Filter contigs for Prokka (≥500 bp)

Some downstream tools struggle with very short contigs or very large numbers of contigs.

In Galaxy: 1. Search for the tool **Filter FASTA** (it may appear as *Filter sequences by length*). 2. Input: your original SPAdes **contigs FASTA**. 3. Set **minimum sequence length = 500 bp**. 4. Run the tool to create a **filtered contigs** dataset.

Use the filtered contigs for Prokka today.

Keep the original (unfiltered) contigs for TYGS. TYGS can take a long time and you already submitted the full assembly there.

0C. Filtered contigs dataset name (Galaxy History):

31: Filter FASTA on dataset 27: FASTA sequences

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1) Genome QC (CheckM2 in Galaxy)

Today you will start by evaluating your assembly quality in **Galaxy** using **CheckM2**, which estimates **completeness** and **contamination**.

1A — CheckM2

In Galaxy: 1. Run **CheckM2** on the **filtered assembly FASTA** you will use for Prokka. 2. Open the main CheckM2 results table/report.

CheckM2 tool version (from Job Info):

CheckM2 Galaxy Version 1.1.0+galaxy0

Completeness (%):

98.96

Contamination (%):

0.1

Interpretation (2–3 sentences): Would you trust this assembly for annotation/species ID? Why?

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2) Run Prokka on Galaxy

In Galaxy: 1. Search tools for **Prokka** 2. Input: your SPAdes **contigs FASTA** 3. Use defaults unless instructed otherwise 4. Run Prokka

Record your Prokka settings

2A. Prokka version (from Job Info):

Prokka Galaxy Version 1.14.6+galaxy1

2B. Any settings changed from defaults? If yes, what?

Force GenBank compliance- YES

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3) Prokka outputs — what is each file for?

From Prokka outputs in Galaxy, you will use these most today:

- **.txt** summary statistics
- **.gff** master annotation (features + coordinates)

- **.ffn** nucleotide sequences of gene features (includes rRNA genes)
- **.faa** amino acid sequences of predicted proteins

3A. In your own words, what is the difference between `.gff`, `.ffn`, and `.faa`? (2–4 sentences)

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4) Interpret the Prokka summary (.txt)

Open the Prokka **.txt** summary.

4A. Organism line (paste exactly as reported — it may be "Unknown"):

Genus species strain

4B. contigs:

79

4C. bases:

4797060

4D. CDS:

4507

3E. rRNA (total rRNA features):

3

3F. tRNA:

62

3G. tmRNA:

1

3H. Based on genome size and CDS count, does this look like a typical bacterial genome? Why/why not? (2–3 sentences)

Yes it does look like a typical bacterial genome.

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5) rRNA features (16S/23S/5S) — paste evidence from Prokka

The Prokka `.txt` summary often reports only a total count (e.g., **rRNA: 3**) without listing 16S vs 23S vs 5S in that summary.

To see which rRNAs were found: - Open the **.gff** file and search for `rRNA`, `16S`, `23S`, `5S` **OR** - Open the **.ffn** file and search for `ribosomal RNA`

5A. Paste the description line (header or attribute text) for the 16S rRNA feature:

gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - .
ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal

5B. Paste the description line (header or attribute text) for the 23S rRNA feature:

ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)

5C. Paste the description line (header or attribute text) for the 5S rRNA feature:

RNAgnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + .
ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal

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6) Hypothetical proteins — estimate and interpret

Hypothetical proteins typically appear as: - `product=hypothetical protein` in `.gff` - "hypothetical protein" in headers/annotations in `.faa`

6A. How did you estimate or count hypothetical proteins? (1–2 sentences)

Command f and typed in hypothetical

6B. Estimated number of hypothetical proteins:

1098

6C. Estimated % hypothetical proteins (of CDS):

24.01%

6D. What does a high % hypothetical proteins mean (and NOT mean)? (2–3 sentences)

7) Extract 16S rRNA from the Prokka .ffn

Open the Prokka **.ffn** file. Search for a header/annotation indicating **16S ribosomal RNA**.

7A. Paste the FASTA header line for the 16S entry you used:

>NCJPFIPA_04563 16S ribosomal RNA

7B. Paste the full 16S rRNA sequence you used for BLAST (sequence only; you can keep line breaks):

```
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

7C. In 1 sentence: why do we use 16S for a quick ID?

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8) 16S BLAST (species ID quick check)

BLAST your 16S sequence: 1. Go to NCBI BLAST 2. Choose **BLASTn** 3. Paste your 16S sequence 4. Run BLAST and choose a reasonable hit to interpret

Upload screenshot: your 16S BLAST results (Descriptions table or Alignment view)
Choose File no file selected

8A. Top hit organism name (as shown):

Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence

8B. % identity and query coverage for that hit:

99.48% identities and 100% Query Cover

8C. Your best interpretation of species/genus (be cautious) (2–3 sentences):

8D. One reason 16S BLAST can be misleading or insufficient (1–2 sentences):

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9) ANI confirmation (EZBioCloud ANI)

After you have a likely species ID from **16S BLAST**, use ANI as a stronger genome-wide comparison.

You will use the **EZBioCloud ANI** tool here (open in a new tab): https://www.ezbiocloud.net/tools/ani (https://www.ezbiocloud.net/tools/ani)

What you will do 1. Download your **FILTERED contigs FASTA** (the same one you used for Prokka today). 2. Download **two reference genomes** (FASTA) from NCBI for *two candidate species* you want to compare against (often the top 16S BLAST hit and one close alternative). 3. Run ANI twice (your genome vs reference genome 1; your genome vs reference genome 2).

9A. Species/strain for comparison #1 (your chosen reference):

e.g., Genus species strain (accession)

9B. ANI for comparison #1 (%):

(your ANI %)

9C. Species/strain for comparison #2 (your chosen reference):

e.g., Genus species strain (accession)

9D. ANI for comparison #2 (%):

(your ANI %)

9E. Interpretation (2–3 sentences): Which comparison supports the strongest species-level match, and why?

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10) TYGS — what it provides

Submit your assembly to TYGS (Type Strain Genome Server). TYGS will: - Compare your genome to type strains - Provide **dDDH** (species boundary ~70%) - Produce a phylogenomic tree

10A. In your own words: what is TYGS doing that 16S BLAST is not? (2–4 sentences)

10B. Record your closest type strain match (name) and dDDH value (if available today):

Paste name + dDDH if available

10C. What does the 70% dDDH threshold mean? (1–2 sentences)

10D. If TYGS shows low branch support and offers a proteome-based tree, what does that imply? (2–3 sentences)

Hint: same genomes, but tree inferred from predicted proteins to improve phylogenetic signal/support.

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11) iTOL — visualize your TYGS tree

1. From TYGS, download the **tree file** (**.phy**)
2. Go to **iTOL** (Interactive Tree of Life)
3. Upload the tree
4. Adjust labels/formatting enough to be readable
5. Take a screenshot

Upload screenshot: iTOL tree visualization (from TYGS tree)
Choose File no file selected

11A. What does the tree tell you that a single BLAST hit does not? (2–3 sentences)

11B. Identify one branch you trust and one you would be cautious about. Explain using support values if shown. (2–4 sentences)

12) What to download and save for Git

Minimum recommended downloads for your local folder: - Prokka **.txt** summary - Prokka **.gff** - Prokka **.ffn** - Prokka **.faa** - CheckM2 results/report (table) - TYGS tree file (**.phy**) - Your PDF from this page (print-to-PDF)

12A. Write 4–6 lines of a "workflow note" you would put in `workflow_day13.md` (include tool versions when possible).

Example: Prokka vX on Galaxy; input contigs; outputs saved; extracted 16S from .ffn; BLASTn; ran CheckM2 (completeness/contamination); submitted assembly to TYGS; downloaded tree; visualized in iTOL...

Export (PDF)

1. Upload the required screenshots above (BLAST + iTOL)
2. Click **Prepare submission preview**
3. Click **Download as PDF (Print)**

Submission preview

Setup
Assembly file: 27: SPAdes on dataset 15 and 16 and collection 12: Contigs
Local folder: Bioinformatics/biol4810/module02/Annotation/

Prokka run
Version: Prokka Galaxy Version 1.14.6+galaxy1
Settings: Force GenBank compliance- YES

Outputs understanding
.gff vs .ffn vs .faa: -

Prokka summary (.txt)
Organism: Genus species strain
Contigs: 79 *Bases:* 4797060

CDS: 4507 *rRNA:* 3 *tRNA:* 62 *tmRNA:* 1
Sanity check: Yes it does look like a typical bacterial genome.

rRNA evidence
16S description: gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - . ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal
23S description: gnl|Prokka|NCJPFIPA_53 barrnap:0.9 rRNA 4 1724 0 + . ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
5S description: RNAgnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + . ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal RNA

CheckM2
CheckM2 version: CheckM2 Galaxy Version 1.1.0+galaxy0
Completeness: 98.96% *Contamination:* 0.1%
CheckM2 interpretation: -

Hypothetical proteins
How estimated: Command f and typed in hypothetical
 # hypothetical: 1098 *% hypothetical:* 24.01%
Interpretation: -

16S extraction
16S header: >NCJPFIPA_04563 16S ribosomal RNA
16S sequence:

```
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

Why 16S: -

16S BLAST

Top hit: Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence

Metrics: 99.48% identities and 100% Query Cover

Interpretation: -

Limitation: -

BLAST screenshot

(no image uploaded)

TYGS

TYGS vs 16S: -

Closest match + dDDH: -

70% threshold: -

Proteome tree option (low support): -

iTOL tree

Tree adds value because: -
Support/trust: -

iTOL tree screenshot

(no image uploaded)

Workflow note (for Git)

```
-
```